INCH-POUND

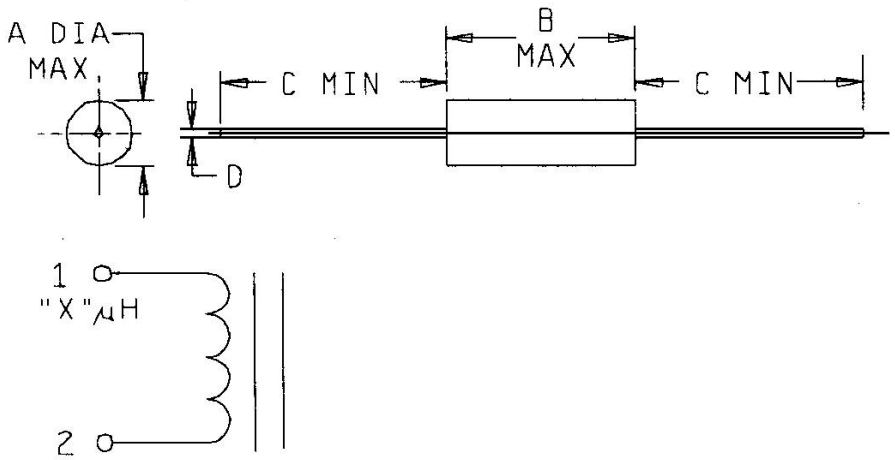
MIL-PRF-27/286C
1 October 2010
SUPERSEDING
MIL-PRF-27/286B
5 April 1990

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS AND INDUCTORS
(AUDIO, POWER AND HIGH-POWER PULSE)
INDUCTORS, POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.




CIRCUIT DIAGRAM AND MARKING

| Dimensions | |
|---|---|
| A | .375 (9.53) |
| B | .500 (12.7) |
| C | 1.25 (31.8) |
| D | .032 (0.81) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Marking shall be on the side of the case.
5. Electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations.



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REQUIREMENTS:

Electrical ratings: See table I.

TABLE I. Electrical ratings.

| Dash no. 1/ | Inductance µh (1-2) +30 percent -5 percent 2/ 3/ | DC current amps (1-2) max 3/ 4/ | DC resistance ohms (max) (1-2) 3/ | Voltage (1-2) 3/ |
|---|---|---|---|---|
| 01 | 120 | 1.6 | 0.14 | 1 volt 20 kHz |
| 02 | 300 | 1.0 | 0.35 | 1 volt 20 kHz |
| 03 | 500 | 0.75 | 0.56 | 1 volt 20 kHz |
| 04 | 1200 | 0.50 | 1.40 | 1 volt 20 kHz |
| 05 | 3000 | 0.30 | 3.50 | 1 volt 20 kHz |
| 06 | 5000 | 0.25 | 5.60 | 1 volt 20 kHz |
| 07 | 30 | 3 | .035 | 1 volt 20 kHz |
| 08 | 50 | 2.5 | .056 | 1 volt 20 kHz |

1/ Qualification test and approval to M27/286-06, shall be sufficient to grant qualification approval to M27/286-01 through M27/286-08.
2/ The inductance is measured with 0 A dc applied to (1-2) and at the specified voltage across (1-2).
3/ When numbers are in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.
4/ At rated dc current, inductance will exceed 80 percent of rated inductance.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Epoxy.

Terminals: .032 inch oxygen free copper leads.

Weight: 4.5 grams, maximum.

Altitude: 75,000 feet, maximum.

Operating temperature range: -55°C to +105°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.



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Dielectric withstanding voltage:

At sea level: 100 volts rms.

At reduce barometric pressure: 100 volts rms.

Vibration (high frequency): MIL-STD-202, method 204.

Temperature rise: +45°C with 1 volt rms, 20 kHz, current as specified in table I applied to (1-2) at an ambient temperature of +60°C maximum.

Marking location: See figure 1.

Part or Identification Number (PIN): M27/286-(dash number from table I).

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents.

MIL-PRF-27
MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2009-066)

Review activities:
Army - AR, MI
Navy - AS, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.



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